

Data-science for scientific R&D course

- 0) As a first step, please create a dedicated folder to store all contents associated with the current workshop, say 'ds_mini_course' within Documents or Downloads folder on your PC. If possible, avoid using cloud-based storage services such as Box, OneDrive, etc.

- 1) To install miniconda

Option 1: Quick install via command line (ETI < 1min per step)

<https://docs.conda.io/projects/miniconda/en/latest/index.html#quick-command-line-install>

OR

Option 2: Via graphical installer

<https://docs.conda.io/projects/miniconda/en/latest/miniconda-install.html>

- 2) To create a new environment in conda and install python (ETI < 1min per step):

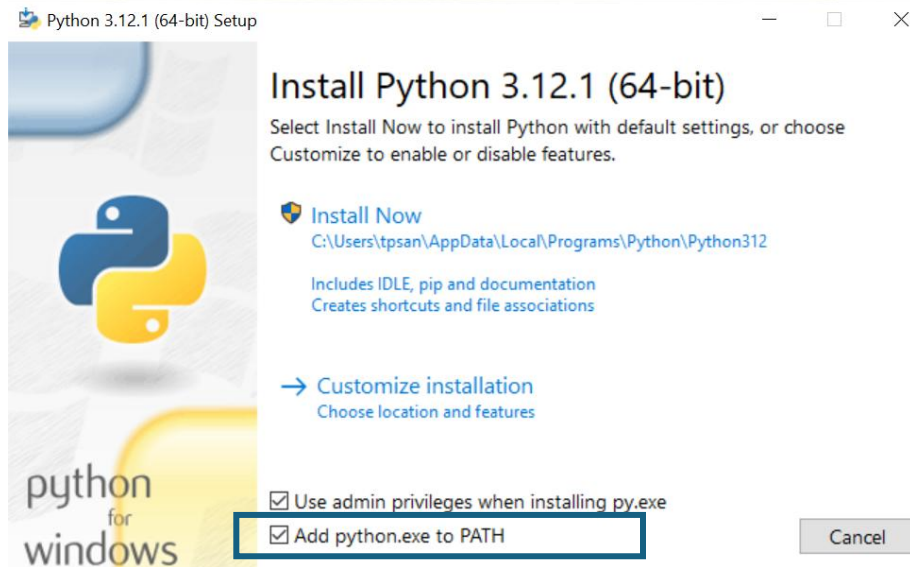
```
conda create --name ds_mini_course python=3.12  
conda activate ds_mini_course
```

- 3) To install python IDLE (Integrated Development and Learning Environment) [ETI ~ 3 min]:

Go to <https://www.python.org/downloads/>

Download the OS specific installer

Make sure to opt-in for “Add python.exe to PATH”



- 4) To install VSCode (Optional but recommended for intermediate/ advanced python users):

<https://code.visualstudio.com/>

- 5) To install python libraries:

It is always recommended to install packages in a project specific environment as against the default base environment.

```
conda activate ds_mini_course
```

Replace <package_name> with desired library name while running the following command

```
conda install -c conda-forge <package_name>
```

Conda-forge channel enables quicker installation of packages. If the above step fails, one could try the default conda installation using

```
conda install <package_name>
```

(ETI ~ 1 min per package using conda-forge channel)

<package_name> = scikit-learn (python library for machine learning)

<package_name> = pandas (python library for processing data-sets)

<package_name> = matplotlib (python library for plotting)

<package_name> = numpy (python library for numerical operations and scientific computing; Optional as it gets installed by default with scikit-learn)